BIFX 546 Midterm Presentation — Supporting Document

Predicting Parkinson's Disease Severity from Voice Features: EDA & Preliminary Findings

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Dataset: Parkinsons Telemonitoring (UCI Machine Learning Repository)

Executive Summary:
This document accompanies the Midterm Presentation (14 slides, ~6 minutes) and provides detailed context, findings, and next steps for the Parkinsons Telemonitoring project. The presentation focuses on exploratory data analysis (EDA) findings and establishes the foundation for the modeling phase leading to the Final Demo.

Key Finding:
Voice features show moderate but meaningful correlation with Parkinson's disease severity (UPDRS scores), supporting the feasibility of voice-based prediction. However, high multicollinearity among features requires careful regularization and feature selection strategies in the modeling phase.

1. Research Question & Motivation

Central Question
Can we predict Parkinson's disease severity (UPDRS scores) from home-recorded voice measurements?
Why This Matters

- **Clinical significance:** Early detection and accurate severity tracking enable timely intervention and improved patient outcomes
- **Practical benefit:** Remote telemonitoring via smartphone recordings reduces barriers to care for patients in underserved areas
- **Scientific rationale:** Parkinson's disease affects motor control of the vocal apparatus; voice features may serve as objective biomarkers of disease progression

Background Context
Parkinson's disease is a progressive neurological disorder characterized by motor symptoms (tremor, rigidity, bradykinesia) and non-motor features. Speech impairments (dysarthria) are common and affect approximately 75% of patients. Voice analysis offers:

- Non-invasive, scalable assessment modality
- Quantitative, objective measurements (vs. clinical rating scales)
- Potential for continuous home monitoring

2. Dataset Overview & Characteristics

Source & Basic Statistics:

Attribute	Value
Dataset Name	Parkinsons Telemonitoring
Source	UCI Machine Learning Repository
Sample Size	5,875 voice recordings
Unique Patients	42 subjects

Recording Duration	~6 months per patient (~140 recordings per patient)
Features	22 total (demographics + 16 voice measures + 2 targets)
Data Quality	No missing values, no duplicates

Feature Overview

Demographics (3 features):

- subject# — Patient identifier (1–42)
- age — Patient age at baseline
- sex — Gender (0 = male, 1 = female)
- test_time — Days since recruitment

Target Variables (2 features):

- motor_UPDRS — Unified Parkinson's Disease Rating Scale, motor subscale (0–108, higher = more severe)
- total_UPDRS — Total UPDRS score (0–176, higher = more severe)

Voice Features (16 features): Jitter (%), Jitter: RAP, Jitter:DDP, Shimmer, Shimmer:APQ3, Shimmer:APQ5, Shimmer:APQ11, Shimmer:DDA, NHR, HNR, RPDE, DFA, PPE

3. Summary Statistics: Key Findings

Motor UPDRS (Primary Target)

Statistic	Value	Interpretation
Mean	21.30	Central tendency (mild-to-moderate severity)
Median	20.90	Slightly left of mean (minor right skew)
Std Dev	8.13	Substantial variability (good for modeling)
Min	5.04	Lowest severity in cohort
Max	39.51	Highest severity (advanced motor symptoms)
Range	34.47	Widespread prevents floor/ceiling effects

Total UPDRS (Comprehensive Severity)

Statistic	Value	Interpretation
Mean	29.02	Moderate overall disease burden
Std Dev	10.70	Greater variability than motor subscale alone
Min	7.00	Minimal symptom burden
Max	54.99	Substantial multi-domain impairment

Demographics

Variable	Value	Notes
Age (years)	64.8 ± 12.1	Typical Parkinson's patient age; range 36–85
Gender	68% M / 32% F	Slight male predominance (expected in PD)
Recording Period	−4 to 215 days	Range relative to recruitment; average ~140 per patient

Implication: Adequate variability in target variables supports regression modeling. No extreme floor/ceiling effects. Demographic distribution is clinically reasonable.

4. EDA Findings (Visualization)

Finding #1: Motor UPDRS Distribution (Slightly Right Skewed)

Observation:
The histogram with KDE overlay reveals a slightly right-skewed distribution of Motor UPDRS scores, with most recordings clustered between 15–28 on the 0–108 scale.

Statistics:

- Mode: ~20–22 (most frequent range)
- Mean: 21.30, Median: 20.90
- Skewness: +0.45 (mild right skew)

Interpretation:

- Central tendency in mild-to-moderate range reflects the cohort's early-stage disease status
- Right skew indicates a tail of more severely affected patients (expected; PD is progressive)
- Adequate spread prevents modeling artifacts (floor/ceiling)

Clinical Context:
Early-stage PD patients have milder motor deficits. The distribution captures this well, with sufficient representation of both less severe and more symptomatic cases.

Finding #2: Voice Features Correlate Moderately with UPDRS ($r \approx 0.15$–0.35)

Observation:
Correlation heatmap (voice features × UPDRS scores) reveals moderate but statistically significant relationships between specific voice measures and disease severity.

Key Correlations:

Voice Feature	Correlation with Motor UPDRS	Interpretation
PPE (pitch period entropy)	+0.25 to +0.35	Moderate positive — higher PPE → more severe
Jitter (%)	+0.15 to +0.22	Weak-to-moderate — pitch instability marker
Shimmer	+0.12 to +0.18	Weak — amplitude variation marker
HNR (harmonic-to-noise ratio)	−0.16	Slight negative — lower HNR → noisier voice
RPDE (recurrence period density entropy)	+0.20	Weak-to-moderate — complexity/irregularity
DFA (detrended fluctuation analysis)	+0.18	Weak — fractal dimension marker

Important Note: Motor UPDRS and Total UPDRS are highly correlated ($r \approx 0.95$), as expected (motor subscale is major component of total).

Interpretation:

- Voice features **do contain signals** related to disease severity.
- Correlations are **moderate, not strong** → suggests non-linear relationships or that voice alone is partial predictor.
- Multiple features contribute → ensemble modeling (e.g., regression, random forest) likely more effective than single feature prediction.

Clinical Significance:
Parkinson's motor pathology affects vocal tract motor control (laryngeal, articulatory muscles). The observed correlations support this mechanism — PPE and jitter, which measure pitch stability/irregularity, reflect motor dysfunction.

Finding #3: High Multicollinearity Among Voice Features (r > 0.80 for similar measures)

Observation:
Voice features exhibit strong pairwise correlations, particularly among variants measuring the same underlying phenomenon.
Examples of Multicollinearity:

Feature Pair	Correlation	Why Correlated
Jitter(%) ↔ Jitter:RAP	r = 0.91	Both measure pitch variation; different calculation methods
Jitter(%) ↔ Jitter:DDP	r = 0.88	Jitter variants; DDP = relative average perturbation
Shimmer ↔ Shimmer:APQ3	r = 0.83	Both measure amplitude variation at different lag times
Shimmer ↔ Shimmer:DDA	r = 0.85	Amplitude perturbation measures

Correlation Structure:
- Jitter variants highly intercorrelated (r > 0.80)
- Shimmer variants are highly intercorrelated (r > 0.85)
- Cross-cluster correlations moderate (r ≈ 0.40–0.60)
- Suggests ~3–4 underlying dimensions (pitch stability, amplitude variation, noise level, complexity)

Modeling Implication:
Standard OLS regression with all sixteen voice features will suffer from:
- **Unstable coefficient estimates** (slight changes in data → large coefficient swings)
- **Inflated standard errors** (wider confidence intervals, unreliable inference)
- **Overfitting risk** (model learns collinear noise rather than signal)

Mitigation Strategies for Final Demo:
1. **Regularization (Ridge/Lasso):** Penalizes coefficient magnitude, stabilizes estimates.
2. **Feature Selection:** Retain one representative jitter measure + one shimmer measure (reducing redundancy)
3. **Dimensionality Reduction (PCA):** Extract uncorrelated principal components.
4. **Tree-based Models:** Random Forest / Gradient Boosting are less sensitive to multicollinearity.

Decision: Will implement Ridge/Lasso regression + compare with single-feature-per-category baseline.

Finding #4: Disease Progression Over 6 Months (Upward Trend in UPDRS)

Observation:
Time-series plots (Motor UPDRS vs. test_time for 5 sampled patients) reveal a general upward trend in UPDRS scores over the ~6-month observation window.
Pattern:
- **Patient 1:** UPDRS increases from ~18 → ~26 (gradual slope)
- **Patient 2:** UPDRS increases from ~15 → ~22 (steeper slope)
- **Patient 3:** UPDRS increases from ~20 → ~28 (moderate slope)
- **Patient 4:** UPDRS increases from ~10 → ~18 (steeper slope)
- **Patient 5:** UPDRS increases from ~25 → ~32 (consistent progression)

Group-Level Trend:
Average slope ≈ +0.05 UPDRS points per day (~+10 points over 6 months), consistent with early-stage PD progression rates reported in literature.
Interpretation:
- **Validates dataset integrity:** Disease progression matches clinical expectations.

- **Confirms repeated-measures structure:** Each patient has temporal trajectory (justifies patient-level analysis)
- **Indicates modeling complexity:** Standard regression assumes independence; repeated measures need careful handling.